

The Data Layer

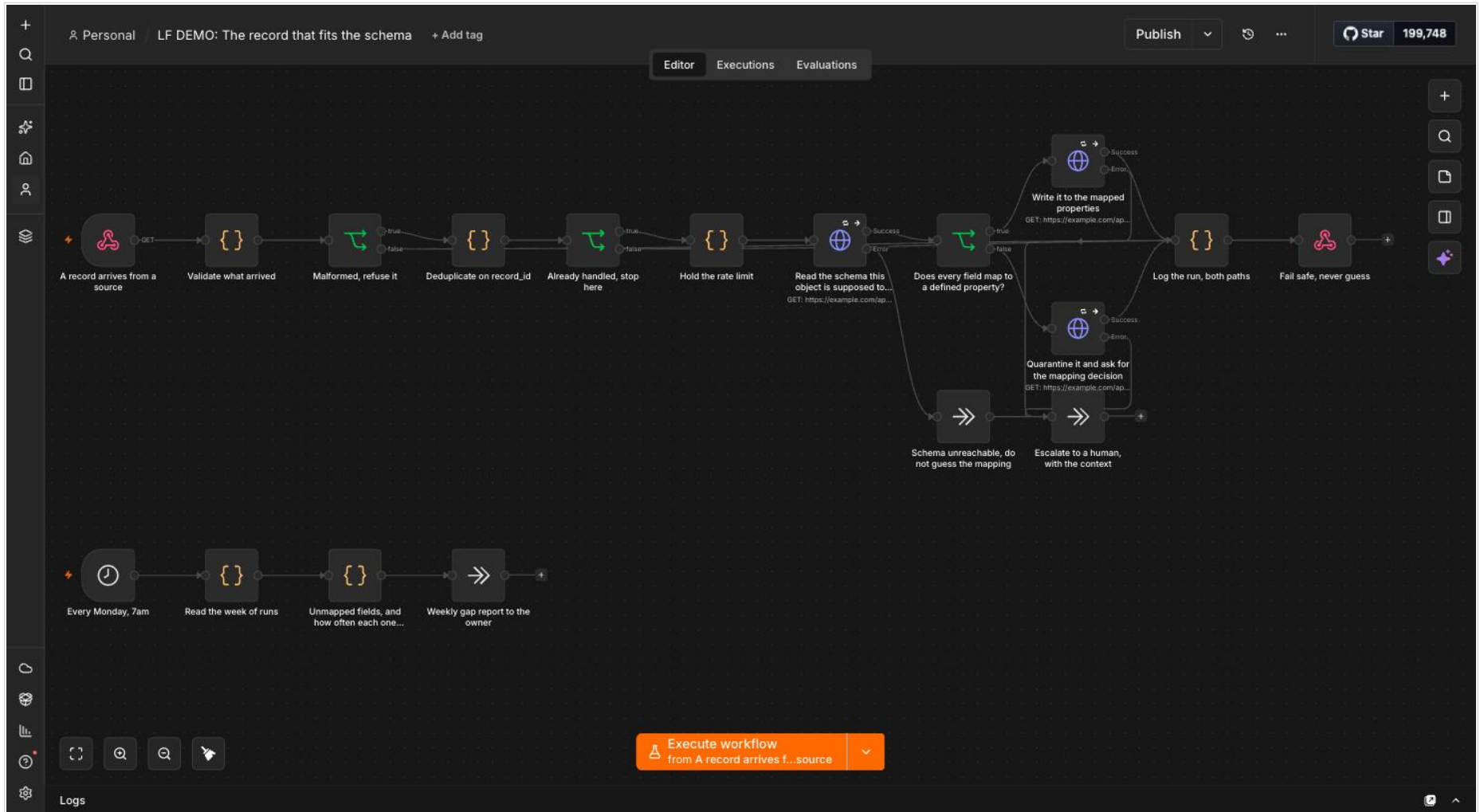
data-model-architecture · built in n8n

Most CRM problems are data-model problems wearing a workflow costume.

What it connects to: The schema has to hold for every system that writes into it. A property that only makes sense in one tool is the one that breaks the reporting later.

1 of 3 SYSTEM

Here's the system that enforces it — a record only gets written once every field maps to a property that exists.



2 of 3 THINKING

This is the gate that keeps the model clean: every field has to map to a property that exists, or the record is quarantined rather than guessed at. Guessing is how you end up with three fields meaning the same thing.

The screenshot shows a dark-themed user interface for a data processing tool. The main window is titled "Does every field map to a defined property?". The interface is divided into two main sections: "INPUT" on the left and "Parameters" / "Settings" on the right.

INPUT Section:

- At the top, it says "INPUT".
- Below that, there is a large area with a "No input data" message and a "Execute previous nodes" button.
- At the bottom, there is a "to view input data" link.

Parameters / Settings Section:

- There are two tabs: "Parameters" (selected) and "Settings".
- Under the "Parameters" tab, there is a "Conditions" section.
- The "Conditions" section contains a single condition: `{{ $json.fullyMapped === true && $json.objectExists === true }}`. To the right of the code is a checkbox labeled "is true" which is checked.
- Below the condition is a "+ Add condition" button.
- Below that is a toggle switch labeled "Convert types where required", which is currently turned off.
- Below the toggle is an "Options" section with a "+ Add option" button.
- At the top right of the "Parameters" tab, there is an orange button labeled "Execute step".

And the property architecture underneath it, with fill rates so you can see what's actually used.

Unmapped fields, and how often each one arrived

INPUT

→|

{} No input data

Execute previous nodes

to view input data

Parameters Settings

Execute step

Mode

Run Once for All Items

Language

JavaScript

JavaScript

Code Ask AI

```
1 // Unmapped fields, and how often each one arrived – the schema gap,
2 ranked.
3 const recs = Object.values($getWorkflowStaticData('global').records || {});
4 };
5 const week = recs.filter(r => r.at > Date.now() - 7*864e5);
6 const counts = {};
7 for (const r of week) for (const f of (r.unmapped || [])) counts[f] =
8 (counts[f] || 0) + 1;
9 const ranked = Object.entries(counts).sort((a, b) => b[1] -
10 a[1]).slice(0, 5);
11 const quarantined = week.filter(r => r.quarantined);
12 return [{ json: {
13   received: week.length,
14   quarantined: quarantined.length,
15   worstFields: ranked,
16   note: ranked.length
17   ? `"${ranked[0][0]}" arrived ${ranked[0][1]} times with nowhere to
18   go. Decide the property once and ${quarantined.length} quarantined
```

Type \$ for a list of **special vars/methods**. Debug by using `console.log()` statements and viewing their output in the browser console.